THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 6: **Measuring Space: Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 7 cm is: 1

(a) 44 cm **(b)** 88 cm **(c)** 22 cm **(d)** 154 cm

2. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 35 cm is: 1

(a) 3850 cm **(b)** 110 cm **(c)** 440 cm **(d)** 220 cm

3. The perimeter of a circle is 66 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 11.5 cm **(b)** 10.5 cm **(c)** 21 cm **(d)** 5.25 cm

4. The length of an arc subtending 90° at the centre of a circle of radius 7 cm ($\pi = \frac{22}{7}$) is: 1

(a) 5.5 cm **(b)** 11 cm **(c)** 22 cm **(d)** 44 cm

5. By Heron's formula, the area of a triangle with sides 5 cm, 12 cm and 13 cm is: 1

(a) 30 cm^2 **(b)** 24 cm^2 **(c)** 36 cm^2 **(d)** 60 cm^2

6. By Heron's formula, the area of a triangle with sides 9 cm, 10 cm and 17 cm is: 1

(a) 90 cm^2 **(b)** 42 cm^2 **(c)** 30 cm^2 **(d)** 36 cm^2

7. The area of a triangle with base 10 cm and height 7 cm is: 1

(a) 70 cm^2 **(b)** 17 cm^2 **(c)** 45 cm^2 **(d)** 35 cm^2

8. By Brahmagupta's formula, the area of a cyclic quadrilateral with sides $5, 5, 5, 5$ is: 1

(a) 29 sq. units **(b)** 23 sq. units **(c)** 25 sq. units **(d)** 50 sq. units

9. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 21 cm is: 1

(a) 2772 cm^2 **(b)** 1386 cm^2 **(c)** 693 cm^2 **(d)** 132 cm^2

10. The length of a semicircle of radius r is: 1

(a) πr^2 **(b)** πr **(c)** $2\pi r$ **(d)** $\frac{\pi r}{2}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The perimeter of a circle is 44 cm. Find its radius ($\pi = \frac{22}{7}$). 2

12. Find, correct to 3 significant figures, the circumference of a circle of radius 10 cm ($\pi = \frac{22}{7}$). 2

13. Find the length of the arc of a circle of radius 3.5 cm if the angle at the centre is 60° ($\pi = \frac{22}{7}$). 2

14. Find the perimeter of a sector of a circle of radius 14 cm and sector angle 75° ($\pi = \frac{22}{7}$). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A track has two straight sections of 84.39 m each and two semicircular ends of radius 36.5 m. Find the length of the innermost edge (use $\pi = 3.1416$). 3

16. The lanes on a circular bend are 1.2 m wide. By how much does the length of a full semicircular bend differ between two adjacent lanes (the stagger needed)? Use $\pi = 3.14$. 3

17. Two circles of radius r each pass through the centre of the other. Find the perimeter of the shape formed by the outer arcs. 3

OR

Semicircle a on diameter PQ is compared with three semicircles b, c, d on adjacent parts of PQ. Show that the two paths from P to Q are equal in length. 3

18. Show, using inscribed and circumscribed hexagons of a circle of radius r, that $3 < \pi < 2\sqrt{3}$. 3

OR

Find the area of a triangle with sides 13 cm, 14 cm and 15 cm and the height on the side of 14 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the area of a parallelogram with base b and height h is bh. What changes if the parallelogram is 'thin'? 5

OR

Prove that the area of a triangle is $\frac{1}{2}bh$ and deduce that a median divides it into two triangles of equal area. 5

20. **Case study:** A 400 m track has two straights of 84.39 m each and two semicircular ends. The innermost semicircle has radius 36.5 m and each lane is 1.22 m wide. A runner keeps 0.3 m from the inner edge. 5

(i) Find her radius on the bends.

(ii) Find the distance she runs in one circuit ($\pi = 3.1416$).

(iii) Find the stagger needed between adjacent lanes.

(iv) Why must the stagger be the same for every pair of adjacent lanes?

OR

(i) Using Heron's formula show that the area of an isosceles triangle with equal sides a and base $2b$ is $b\sqrt{a^2 - b^2}$. 5

(ii) Check with $a = 5, b = 3$ by the formula $\frac{1}{2}bh$.

END OF QUESTION PAPER — Total: 40 Marks

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